

Code Quality & Reusability Assessment

Date	06 July 2026	Project Name	Smart Lender
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Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes/No)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout	Yes	4 spaces standard in Python files
2	Proper File Structure	Logical organization of source files	Yes	Clear split of train, app, static, templates
3	Meaningful Variables	Reflects variable purpose clearly	Yes	e.g. coapplicant_income_log, features_scaled
4	Method Names	Functions are descriptively named	Yes	e.g. preprocess_input(), load_model()
5	Code Comments	Inline comments explaining complex logic	Yes	Detailed math comments on risk flags and SMOTE
6	Modular Design	Split into reusable functions/modules	Yes	Preprocessing isolated in helper functions
7	No Redundant Code	Duplicate or dead code removed	Yes	Clean code structures, unused model trials removed
8	Error Handling	Exceptions handled gracefully	Yes	Checks for missing rdf.pkl, invalid CSV schema

Reusable Components Summary

S.No	Component Name	Language/Tech	Where Reused	Reusability Level
1	preprocess_input()	Python	Both app.py and train.py for preprocessing	High
2	StandardScaler asset	scikit-learn	Fit in training, transformed during live scoring	High
3	XGBoost classifier	XGBoost	Serialized and loaded on flask startup	High